

Article

Adaptive SOC Estimation of Reconfigurable Battery Modules Based on a Hybrid Deep Learning Model

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Abstract

Reconfigurable battery modules can dynamically adjust the connection topology among battery cells, significantly improving the energy utilization efficiency of battery energy storage systems. However, existing state estimation methods focus primarily on individual battery cells. Frequent topology changes cause traditional State of Charge (SOC) estimation algorithms to accumulate large errors due to mismatches in equivalent capacity and internal resistance, making them ineffective for reconfigurable battery modules. To address this limitation, this paper proposes a Gated Recurrent Unit–Transformer architecture for precise SOC estimation in reconfigurable battery modules. The model uses a Gated Recurrent Unit to capture the temporal continuity of electrochemical evolution and employs the Transformer’s self-attention mechanism to analyze discrete topology changes. Experimental results show excellent estimation accuracy across different initial SOC levels, with a mean absolute error as low as 0.085% at full charge and 0.035% at 50% SOC. The architecture demonstrates strong topology self-identification capability and maintains high robustness even under abrupt voltage steps caused by reconfiguration. This method provides accurate and reliable state estimation for large-scale two-layer reconfigurable battery systems while reducing control complexity and improving operational efficiency.

Keywords: reconfigurable battery module; SOC estimation; Gated Recurrent Unit–Transformer; battery storage system; topology mode recognition

1. Introduction

Amid rapid global electrification and the ongoing transformation of energy systems, large-scale integration of renewable energy has become essential for building low-carbon power systems [1–4]. However, renewable energy sources inherently exhibit significant variability and intermittency. This substantially increases the complexity of real-time power balancing in electricity systems. It also complicates the assurance of a long-term, reliable supply. Consequently, achieving a round-the-clock energy supply becomes highly challenging [5,6]. To tackle these structural challenges, battery energy storage systems (BESS) have emerged as crucial components. They are required to be safe, cost-effective, highly efficient, and long-lasting to balance supply and demand while ensuring power supply stability [7–9]. As BESS gain wider adoption in electric vehicles (EVs), hybrid electric vehicles (HEVs), and backup power systems, their development faces considerable challenges [10].

Traditional fixed-topology battery systems struggle to effectively address the evolution of cell-to-cell inconsistencies during long-term operation, which arise from manufacturing



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process variations and differential operating conditions in extreme environments. These imbalances in the degree of degradation among individual cells not only lead to abrupt capacity changes, significantly reducing usable energy, but also shorten the overall service life of the system due to limited maintenance options, thereby increasing operational and maintenance costs [11,12]. Furthermore, faults in individual or multiple cells can severely degrade system efficiency and, in extreme cases, cause total system failure. To address these issues, reconfigurable battery systems (RBSs) have attracted increasing attention. By incorporating power electronic switches, this design enables real-time isolation and bypassing of degraded or faulty cells through advanced control algorithms. Such reconfigurability is regarded as a key enabler for enhancing system resilience and extending battery service life. Moreover, it offers a promising pathway toward cost-effective maintenance and sustainable BESS.

With the proposition and advancement of the dual-layer structure for RBS [13–15], precise state estimation for reconfigurable battery modules (RBMs) incorporating power switching devices has become essential for achieving efficient and precise control in large-scale systems. In RBMs, these switching devices not only serve as the core physical components enabling topological reconfiguration but also exert a significant influence on the overall electrical behavior of the system through their inherent electrical properties and dynamic switching characteristics [16]. On the one hand, the non-negligible on-state resistance of the switches introduces additional voltage drops and power losses, directly affecting the accuracy of terminal voltage and current measurements at the module level. On the other hand, frequent switching of device states during operation leads to real-time changes in the module's connection topology (series, parallel, or hybrid series–parallel configurations), thereby imparting pronounced nonlinear and dynamic characteristics to the battery module [17,18]. Although reconfigurable battery systems provide enhanced flexibility and fault tolerance, frequent topology switching introduces abrupt voltage transients, signal discontinuities, and highly nonlinear dynamic behavior, which significantly increase the difficulty of accurate SOC estimation. Conventional model-based and data-driven methods often exhibit degraded robustness under dynamically reconfigurable operating conditions. Moreover, the dynamic nature of topology reconfiguration significantly complicates state estimation. Inadequate capture of switching transients and induced discontinuities in the estimation models results in considerable cumulative SOC errors at the module level. This impairs active balancing effectiveness and ultimately jeopardizes the operational safety of large-scale energy storage systems in real-world scenarios [19]. It is worth noting that accurate battery state estimation plays an important role in renewable-energy-based microgrids, distributed energy systems, and intelligent energy management applications [20]. Meanwhile, dynamically reconfigurable battery modules equipped with topology-aware SOC estimation capability have the potential to further improve operational flexibility and energy utilization efficiency in advanced renewable energy storage scenarios.

Current research on SOC estimation methods is relatively well-developed and can be categorized into four main types: ampere-hour (AH) integration, open-circuit voltage methods, model-based estimation techniques, and data-driven approaches [21–27]. These methods have been extensively developed primarily for individual cells and are generally not directly applicable to state estimation in RBMs, where frequent topology changes introduce additional complexities. Several studies have explored SOC estimation for battery packs with fixed configurations. For instance, Qiu et al. [28] combined spatiotemporal graph convolutional networks with deep learning to develop a high-accuracy SOC estimation method for battery packs in series and parallel configurations. While the method effectively captures state variations in fixed series and parallel topologies, it does not account for switching effects. Couto et al. [29] developed a partitioned UKF-based method utilizing

sensor networks for distributed state estimation of battery cells in RBS. Nevertheless, the technique is restricted to individual cell-level estimation and does not address dynamic topology changes in RBMs. Ma et al. [30] developed an EKF-based technique for state estimation in reconfigurable battery packs, with reported estimation errors below 2%. Nevertheless, this method focuses solely on individual cell-level estimation within the reconfigurable battery system and fails to consider switching effects or the holistic module-level dynamics. Nima et al. [31] presented a dual Kalman filter (DKF) technique for online estimation of open-circuit voltage and internal resistance in reconfigurable battery packs. Although the method provides reasonably accurate pack-level information, it fails to sufficiently consider the influence of switching and cannot handle variations in module topology. Nonetheless, it offers a valuable reference for subsequent research. In summary, while existing approaches demonstrate strong performance in conventional or fixed-configuration battery systems, they generally fall short in addressing the unique challenges posed by frequent topology reconfiguration and switching dynamics inherent to RBMs. This gap highlights the need for tailored estimation strategies that explicitly incorporate these effects. In addition to state estimation, deep learning has demonstrated significant potential in the control and optimization of battery energy systems. Recent studies have explored deep reinforcement learning (DRL) for complex decision-making tasks. For instance, a novel AI-driven model-based control framework was proposed to optimize home energy management for solar electric vehicles, demonstrating the power of integrating predictive models with reinforcement learning [32]. While this paper focuses on SOC estimation within reconfigurable topologies, such high-fidelity state information is indispensable as a core input for the aforementioned DRL-based optimization frameworks to ensure system safety and efficiency.

To overcome these limitations and address the challenges posed by signal discontinuities caused by topology switching, as well as the complex nonlinear dynamics of batteries, while accurately estimating the SOC of RBMs containing switching devices, this paper proposes a GRU-Transformer state estimation method. Unlike conventional stationary SOC estimation problems, dynamically reconfigurable battery modules exhibit topology-dependent electrical behavior, abrupt transient responses, and signal discontinuities during switching operations. Therefore, the proposed method is designed as a topology-aware feature learning architecture rather than a generic time-series prediction model. This method integrates the temporal memory capability of Gated Recurrent Units (GRUs) for electrochemical evolution processes with the global attention capture capability of the Transformer architecture for discrete topology logic, constructing an advanced engineering model that combines physical continuity awareness with topology dynamic recognition. Specifically, the GRU component preserves the temporal continuity of electrochemical state evolution and suppresses transient estimation oscillations caused by abrupt voltage variations, while the Transformer component identifies topology-dependent signal discontinuities through the self-attention mechanism. Their synergistic interaction enables effective decoupling between topology-switching disturbances and intrinsic SOC dynamics. Thanks to its superior topology-decoupling capability, the proposed method can effectively eliminate the severe terminal voltage step disturbances induced by RBM reconfiguration and achieve accurate module-level SOC estimation. It should be noted that the primary objective of this work is topology-aware SOC estimation at the module level rather than independent SOC estimation for individual cells. The main contributions of this work can be summarized as follows:

- (1) A topology-aware SOC estimation framework for reconfigurable battery modules is proposed to address signal discontinuity and nonlinear dynamic behavior induced by topology switching.

- (2) The equivalent electrical characteristics of different reconfigurable topologies are systematically analyzed to support topology interpretation and state-space construction.
- (3) Experimental validation under dynamic stress test conditions demonstrates the effectiveness and robustness of the proposed method for SOC estimation in dynamically reconfigurable operating scenarios.

The remainder of this paper is organized as follows. Section 2 introduces the topology of the RBM and provides a modeling analysis of the module topology under different modes. Section 3 describes the structure and implementation process of the proposed method. In Section 4, the experimental and verification results of the proposed method are discussed. Finally, Section 5 delineates the conclusions of the paper.

2. Reconfigurable Modeling and Topological Analysis

To enable a thorough discussion and validation of module state estimation across series, parallel, and series-parallel configurations, the reconfigurable topology employed in this paper primarily utilizes a three-switch configuration. It should be noted that the equivalent circuit modeling presented in this section is primarily intended to analyze the topology-dependent electrical characteristics of RBMs and provide the physical basis for state-space construction in the subsequent learning framework, rather than being directly embedded into the neural network as explicit physics constraints.

2.1. RBM Topological Analysis

With the development of RBS topologies and to accommodate large-scale energy storage systems, a dual-layer reconfigurable topology has been proposed and developed [33,34]. At this stage, the RBS primarily comprises RBMs and switching devices, both of which possess reconfiguration capabilities, as shown in Figure 1, with the RBM regarded as a small-scale independent RBS.

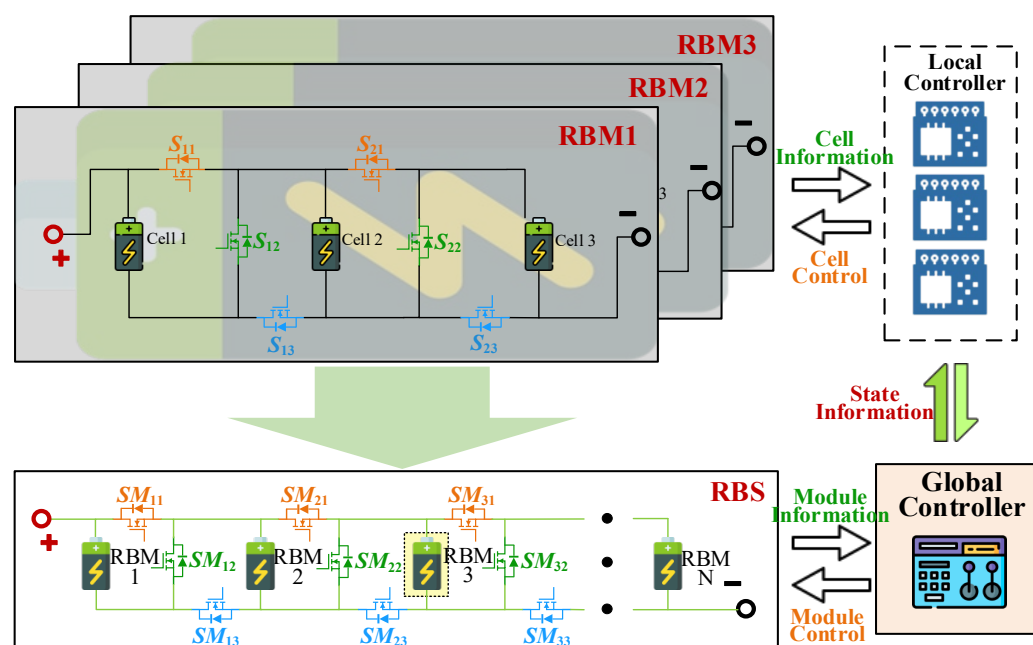


Figure 1. Schematic diagram of master-slave control in double-layer RBS topology.

In this architecture, the connection topology among cells within the module is determined and controlled by electronic switches, enabling real-time reconfiguration of the module's electrical connection mode (series, parallel, or hybrid), as shown in Figure 2. Specifically, switch S_{12} is responsible for controlling series connections between the batteries,

while switches S_{i1} and S_{i3} control parallel connections [35,36]. Consequently, when modeling or analyzing an RBM, its internal structural reconfigurability must be explicitly taken into account. The principal operating modes and corresponding connection topologies of the RBM are illustrated in Figure 3. Specifically, Figure 3a, Figure 3b, and Figure 3c illustrate the energy flow paths and switch conduction states of the adopted RBM under series, parallel, and hybrid series–parallel operating modes, respectively. As shown in Figure 3a, the series configuration primarily exhibits cumulative voltage characteristics, where the terminal voltage is highly sensitive to variations in individual cell voltages. In contrast, the parallel configuration shown in Figure 3b mainly demonstrates current-sharing behavior, resulting in different equivalent impedance and transient response characteristics. Figure 3c further presents the hybrid series–parallel topology, in which voltage accumulation and current redistribution coexist simultaneously, leading to more complex nonlinear dynamic coupling behavior during topology switching.

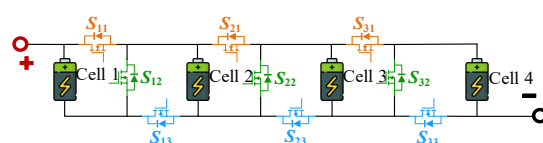


Figure 2. Schematic diagram of RBM topology.

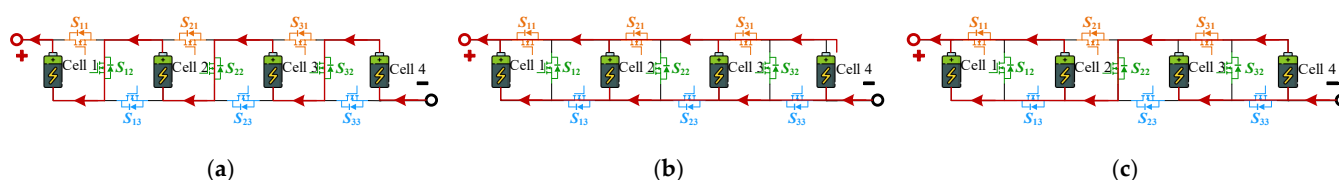


Figure 3. Schematic diagram of RBS operation control. (a) Series connection. (b) Parallel connection. (c) Series–parallel connection.

2.2. Reconfigurable Module Modeling

Accurate SOC estimation for an RBM necessitates the development of a precise equivalent circuit model (ECM). The accuracy of this ECM directly impacts the overall accuracy of the estimation results. Given that the RBM comprises battery cells interconnected through electronic switches, the modeling of the module inherently relies on an accurate equivalent circuit representation of individual cells. Furthermore, owing to the RBM's reconfigurability, the modeling process must explicitly account for the electrical behavior of the module under different connection topologies. For analytical tractability, the derivations in this section assume identical cell parameters and synchronized SOC conditions among battery cells. This simplification is adopted primarily to facilitate the analysis of topology-dependent equivalent electrical characteristics and isolate the influence of topology switching behavior on SOC estimation dynamics.

2.2.1. Battery Equivalent Modeling

Compared to the Rint model and the second-order RC equivalent circuit, the Thevenin model offers a balance of high accuracy and low computational complexity, making it widely adopted in battery balancing and state estimation [37,38]. Its architecture, shown in Figure 4, primarily includes an open-circuit voltage source U_{OC} , an ohmic resistor R_0 , and an RC network comprising a polarization resistor R_1 and a polarization capacitor C_1 . The terminal voltage U_t of the model represents the actual output voltage of the battery. The equivalent circuit derivations and topology-related electrical analyses presented in this section are developed based on conventional battery ECM theory and reconfigurable battery modeling approaches reported in the existing literature [37–39].

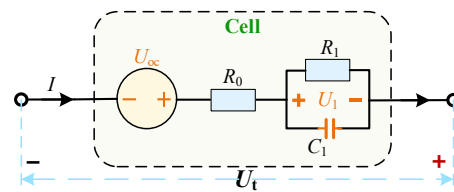


Figure 4. Thevenin equivalent circuit model of a battery.

Based on the electrical characteristics of the ECM, the discrete electrical equations for the battery model are established as follows:

$$U_{1,k} = \exp(-\Delta t/\tau_1)U_{1,k} + R_1(1 - \exp(-\Delta t/\tau_1))I_k \quad (1)$$

$$U_{t,k} = U_{OC,k} - U_{1,k} - R_0I_k \quad (2)$$

where $\tau_1 = R_1C_1$, and Δt is the sampling time. The battery voltage U_t , which includes the polarization voltage U_1 and the open-circuit voltage U_{OC} , serves as the measured signal for SOC estimation. The dynamic SOC of the battery is primarily determined according to Equation (3):

$$SOC_{k+1} = SOC_k - \frac{\eta \Delta t}{3600C_{battery}} \quad (3)$$

where η is the Coulombic efficiency, and $C_{battery}$ is the battery capacity, which can be measured offline or estimated online depending on the application scenario [40].

2.2.2. Equivalent Modeling of Series Mode Modules

In series mode, the N batteries within the module are connected in series, as shown in Figure 3a, to increase the output voltage while maintaining the same capacity. At this time, switch S_{i2} is closed, and all other switches are open. When modeling an RBM with switches, the presence of the switches cannot be ignored. Generally, the impact of the switching devices on the circuit is mainly determined by the total resistance in the conduction path when the switch is on. Therefore, modeling is primarily based on the on-state resistance, denoted as R_S in the figure. For simplicity of analysis, it is assumed that all individual parameters are identical, and the module equivalent model is constructed by aggregating individual battery models. The ECM of the module in series mode is shown in Figure 5.

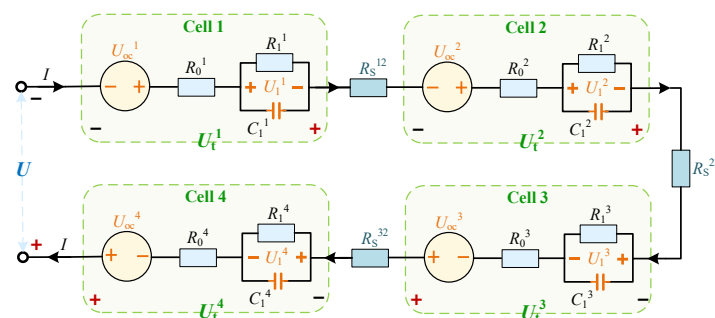


Figure 5. Equivalent circuit diagram in series configuration.

In a series configuration, the terminal voltage U of the RBM is jointly provided by the terminal voltage U_t of the four batteries within the module and the voltage generated by the conduction resistance of the switch:

$$U_k = \sum_{i=1}^N U_{t,k}^i + \sum_{j=1}^{N-1} R_S^{j2} I_k \quad (4)$$

where the current I flowing through each battery is the same. For individual cells with consistent parameters and synchronized SOC, it can be simplified as Equation (5):

$$U_k = N \cdot U_{t,k} + (N - 1)R_S I_k \quad (5)$$

2.2.3. Equivalent Modeling of Parallel Mode Modules

In parallel mode, the N batteries within the module are connected in parallel, as illustrated in Figure 3b, to increase current capacity while maintaining the voltage. Unlike the series state, maintaining the parallel state requires switch S_{i2} to be open, while switches S_{i1} and S_{i3} need to be closed simultaneously. The ECM of the module in parallel mode is shown in Figure 6.

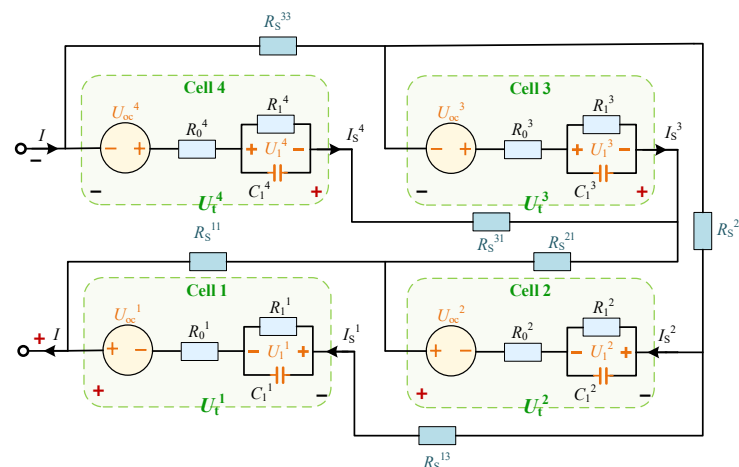


Figure 6. Equivalent circuit diagram in parallel configuration.

At this time, assuming all individual cell parameters are identical and the SOC is initially synchronized, all battery cells share the same terminal voltage U_t . In the meantime, the total current of the module is jointly supplied by the currents of the individual cells:

$$I_k = \sum_{m=1}^N I_{S,k}^m \quad (6)$$

For cells with consistent parameters, they can be equivalently represented as a single monomer model, with parameters scaled proportionally: equivalent ohmic resistance $R_{0,eq} = R_0/N$, equivalent polarization resistance $R_{1,eq} = R_1/N$, and equivalent polarization capacitance $C_{1,eq} = C_1/N$.

2.2.4. Equivalent Modeling of Series–Parallel Mode Modules

The series–parallel configuration represents a composite connection state, primarily achieved through a combination of series and parallel connections. In this architecture, a series–parallel hybrid arrangement typically consists of S parallel groups connected in series to form a series chain. For example, considering a four-cell battery as illustrated in Figure 3c, cells 1 and 2 are connected in parallel to form one parallel group, while cells 3 and 4 constitute another parallel group. These two parallel groups are then connected in series. The specific implementation requires closing switches S_{11} , S_{13} , S_{31} , and S_{33} to form the parallel groups, and then closing S_{22} to connect the two parallel groups in series. The equivalent circuit under this state is shown in Figure 7.

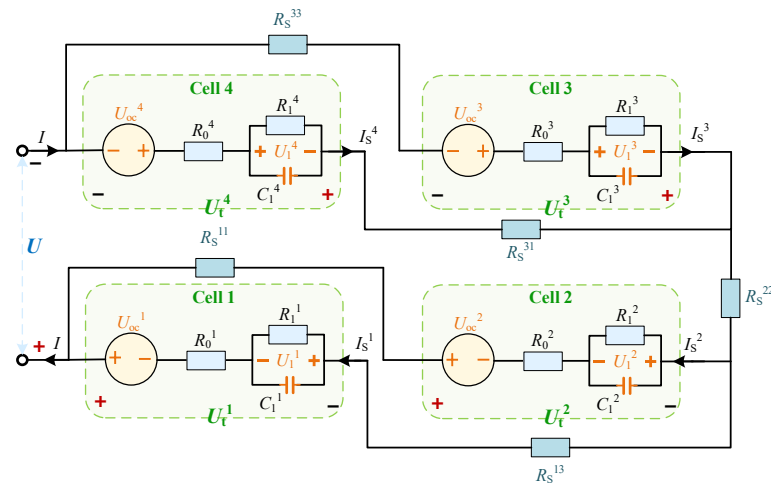


Figure 7. Equivalent circuit diagram of a parallel-series connection.

At this time, the terminal voltage of the module is composed of the terminal voltages of the two parallel groups, while the module current is jointly provided by the currents of the two batteries in the parallel groups.

$$U_k = U_{t,k}^1 + U_{t,k}^3 + R_S^{22} I_k = U_{t,k}^2 + U_{t,k}^4 + R_S^{22} I_k \quad (7)$$

$$I_k = I_{S,k}^1 + I_{S,k}^2 = I_{S,k}^3 + I_{S,k}^4 \quad (8)$$

3. RBM State Estimation Based on GRU-Transformer

In RBMs, the state changes in switches dynamically adjust the internal topological connections of the modules. While this reconfigurability provides operational flexibility, it poses significant challenges for SOC estimation. Different electrical topologies lead to variations in current distribution, terminal voltage measurements, and cell-to-cell interactions, thereby increasing the complexity of accurate SOC prediction. This section introduces a GRU-Transformer hybrid deep learning architecture for SOC estimation in RBMs with switching devices, with a particular focus on the impact of topological variations across different operating modes.

It should be emphasized that the topology information is not merely appended as an additional categorical flag. Instead, it participates in topology-dependent state-space construction and guides the learning of transient feature interactions under dynamically reconfigurable operating conditions. The ECM analysis in Section 2 provides theoretical guidance for defining the topology mode indicator, interpreting voltage transient behavior, and constructing the input state space of the proposed GRU-Transformer architecture. Based on these topology-dependent electrical characteristics, the hybrid network is designed to address signal discontinuities and nonlinear dynamic variations induced by topology switching. In the current study, the proposed framework primarily focuses on topology-dependent electrical dynamics under controlled operating conditions. Thermal dynamics and temperature-dependent electrochemical effects are not explicitly incorporated at this stage in order to isolate the influence of topology switching behavior on SOC estimation performance.

3.1. State Space and Feature Definition

SOC estimation in RBMs essentially involves a nonlinear regression problem driven by dynamic input time series. Given frequent topology reconfigurations, the model must be capable of extracting consistent electrochemical state features from discontinuous and topology-dependent voltage signals. The input feature vector at time t is defined as follows:

$$\mathbf{x}_t = [V_t, I_t, M_t]^T \quad (9)$$

where V_t represents the terminal voltage of the RBM, which is governed by three main factors: the internal resistance of the battery cells, the on-state resistance of the switches, and the open-circuit voltage (OCV) corresponding to the current SOC under the prevailing module topology. I_t denotes the total current of the reconfigurable module, with positive values indicating discharging and negative values indicating charging. $M_t \in \{0, 1, 2\}$ is the discrete topology mode indicator, with the values corresponding to series mode ($M_t = 0$), parallel mode ($M_t = 1$), and hybrid series–parallel mode ($M_t = 2$), respectively.

3.1.1. Sliding Window Sequence Construction

To effectively capture the dynamic lagging effects of the battery's internal states and the integrative nature of the current, this study uses a sliding window technique to transform one-dimensional time-series data into high-dimensional feature representations. With a defined window length L , the input matrix to the model at time t , denoted $\mathbf{X}_{seq}$, is constructed as follows:

$$\mathbf{X}_{seq} = \begin{bmatrix} V_{t-L+1} & I_{t-L+1} & M_{t-L+1} \\ \dots & \dots & \dots \\ V_t & I_t & M_t \end{bmatrix} \quad (10)$$

This temporal construction approach ensures that subsequent GRU layers can access sufficient historical context to model the dynamic evolution logic inherent in AH integration.

3.1.2. Data Preprocessing

To address the distributional differences in terminal voltage V_t and current I_t across different magnitude scales and to prevent neuron saturation during gradient-based training, min-max normalization is applied to the input features as follows:

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (11)$$

where x_{max} and x_{min} denote the maximum and minimum values of V_t and I_t , respectively. V_{max} is defined as the module's maximum safe cutoff voltage, and I_{max} is determined according to the system's maximum allowable discharge current. The topology mode flag M_t is mapped to the $[0, 1]$ interval using label scaling to satisfy the input distribution requirements of the Transformer attention mechanism.

3.2. GRU Layer

3.2.1. Introduction to GRU

While the Transformer architecture effectively captures nonlinear relationships through its powerful global attention mechanism, it lacks an inherent inductive bias for temporal continuity. This makes it challenging to accurately model the physically continuous evolution of SOC over time when applied to RBMs. This limitation becomes particularly evident under operating conditions involving frequent topology mode switching. Abrupt changes in voltage signals often lead to sharp fluctuations in self-attention weights, thereby degrading estimation accuracy. To mitigate this issue, the present work incorporates a GRU as a preceding temporal feature extraction layer.

GRU, a variant of recurrent neural networks similar to LSTM, was developed to address the vanishing/exploding gradient problems in traditional RNNs and to better capture long-term dependencies in sequences. Compared to LSTM, GRU offers a simpler structure,

fewer parameters, and faster training convergence [41]. Its gating mechanism effectively emulates the AH integration process by capturing time-accumulated features of current, thereby imposing physically meaningful constraints on the model. Additionally, GRU provides crucial smoothing and preprocessing of transient voltage and current disturbances caused by RBM topology switches, significantly enhancing the model's robustness and training stability under extreme dynamic conditions.

3.2.2. Implementation of GRU

The GRU controls the flow of information primarily through two key gating mechanisms: the update gate and the reset gate.

The update gate z_t determines the extent to which the previous hidden state h_{t-1} is retained and carried forward to the current hidden state h_t . A higher update gate value results in greater preservation of historical information, while a lower value promotes more substantial incorporation of the new candidate state.

$$z_t = \sigma(\mathbf{W}_z[h_{t-1}, x_t] + b_z) \quad (12)$$

where σ denotes the Sigmoid activation function, which compresses the gating values into the range $[0, 1]$. $\mathbf{W}_z$ and b_z are the learnable weight matrix and bias term, respectively. In the module's state estimation, the update gate can preserve the module's charge memory, preventing drastic fluctuations in the SOC caused by input noise.

The reset gate r_t controls the degree to which past information is incorporated into the computation of the current candidate hidden state $\tilde{h}_t$. In the context of RBMs, abrupt changes in topology lead to sudden shifts in voltage and current characteristics. In such scenarios, the reset gate enables rapid adaptation by assigning a low value to r_t , effectively suppressing voltage-related features related to the previous topology. This mechanism accelerates the model's transition to and convergence within the new topological configuration.

$$r_t = \sigma(\mathbf{W}_r[h_{t-1}, x_t] + b_r) \quad (13)$$

In the implementation of GRU, information constantly transitions between two states:

(1) Candidate Hidden State $\tilde{h}_t$: The candidate hidden state captures newly available information at the current time step through a nonlinear transformation of the input features, which include the module voltage V_t , module current I_t , and discrete topology mode flag M_t .

$$\tilde{h}_t = \tanh(\mathbf{W}[r_t h_{t-1}, x_t] + b) \quad (14)$$

where the tanh function is applied to constrain the candidate hidden state to the range $[-1, 1]$, thereby promoting data centralization and contributing to numerical stability during gradient-based optimization.

(2) Final Output State h_t : The final hidden state of the GRU is computed as a weighted combination of the previous hidden state and the newly computed candidate hidden state, serving as the core update equation that integrates historical and current information in a controlled manner.

$$h_t = (1 - z_t)h_{t-1} + z_t\tilde{h}_t \quad (15)$$

3.3. Transformer Layer

The Transformer is a deep learning architecture that relies solely on the self-attention mechanism. Unlike traditional RNNs, which process sequential data in a strictly temporal and recurrent manner, the Transformer enables parallel processing of all sequence elements, as depicted in Figure 8. In the proposed architecture, the Transformer layer serves as a high-level feature parser, specifically aimed at addressing the limitations of recurrent neural

networks in handling long-range dependencies and complex nonlinear interactions among features. To cope with the non-stationarity of signals caused by topology switching in RBMs, the Transformer dynamically reweights input features through its self-attention mechanism. This allows the model to effectively identify and adapt to patterns associated with abrupt voltage transitions resulting from changes in electrical topology.

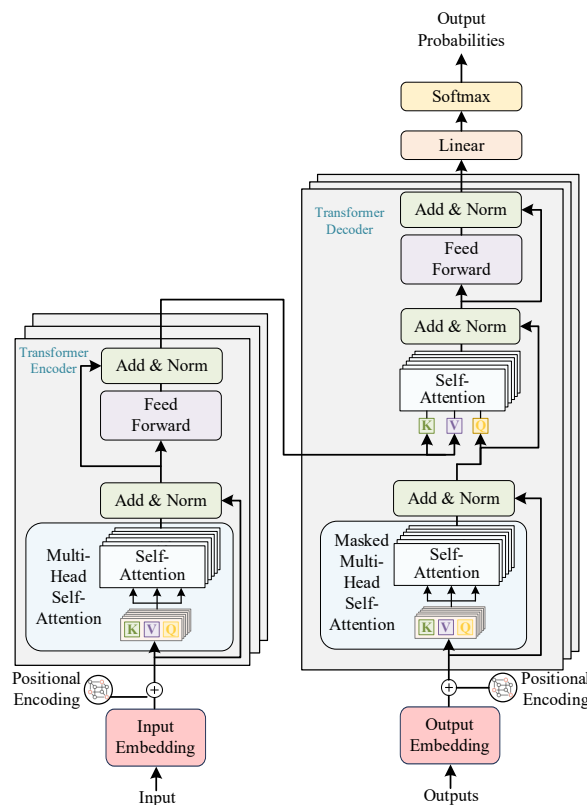


Figure 8. Transformer model architecture diagram.

3.3.1. Positional Encoding

Given that the Transformer architecture is inherently permutation-invariant and lacks an inductive bias for sequential ordering, it is crucial to incorporate positional information into the output vectors $\mathbf{H} = \{h_1, h_2, \dots, h_L\}$ generated by the preceding GRU layer. This step introduces explicit temporal ordering while preserving the parallel computation advantages of the Transformer. In this work, absolute positional encoding based on sine and cosine functions is adopted. This encoding scheme enables the model to effectively capture relative temporal relationships among data points within each sliding window.

$$PE_{(pos,2i)} = \sin(pos/10000^{2i/d_{model}}) \quad (16)$$

$$PE_{(pos,2i+1)} = \cos(pos/10000^{2i/d_{model}}) \quad (17)$$

$$\mathbf{H}_{pos} = \mathbf{H} + PE \quad (18)$$

where pos denotes the time-step index, and i represents the dimension index of the feature vector. The sinusoidal encoding allows the model to capture relative positional relationships through linear transformations, maintaining temporal order during battery degradation. Furthermore, this mechanism enables the model to differentiate between distinct electrochemical evolution stages that may exhibit identical voltage amplitudes at different points in time, which is particularly important for analyzing the dynamic operating conditions of RBMs.

3.3.2. Multi-Head Self-Attention (MHSA)

The MHSA mechanism constitutes the core component of the Transformer's ability to handle mode switching in RBMs. By linearly projecting the input embeddings into h distinct subspaces and performing self-attention independently within each subspace, MHSA allows the model to jointly attend to information from different representation subspaces at different positions. This parallel multi-head processing enables the architecture to capture diverse dependency patterns and contextual relationships across the sequence, which is particularly advantageous for modeling abrupt changes in voltage and current signals induced by topology reconfiguration.

In the single-head self-attention mechanism of the Transformer, input embeddings are linearly projected into three distinct subspaces to produce the query matrix $\mathbf{Q}$, the key matrix $\mathbf{K}$, and the value matrix $\mathbf{V}$. Attention scores are subsequently calculated using the scaled dot-product attention mechanism:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \quad (19)$$

where d_k denotes the dimension of the key vectors, and the scaling factor $\sqrt{d_k}$ stabilizes gradients and prevents excessively large dot-product values.

At this stage, the self-attention mechanism can effectively detect changes in the topology mode indicator M_t and appropriately reduce the influence of anomalous voltage fluctuations in V_t , thus preventing erroneous biases in the predicted SOC. By concatenating the outputs from the h attention heads and applying a subsequent linear transformation, the MHSA layer captures deep and complex dependencies between the topology marker M_t and abrupt voltage jumps observed at the module terminals.

$$\text{MHSA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h)\mathbf{W}^O \quad (20)$$

It is precisely this MHSA mechanism, as formulated in Equation (20), that enables the model to automatically adjust attention weights in response to abrupt terminal voltage changes caused by RBM topology switching. Consequently, the architecture can accurately attribute sudden voltage jumps to topology reconfiguration events rather than power depletion or capacity fade, thus effectively eliminating systematic estimation bias in the predicted SOC.

3.3.3. Feedforward Neural Networks and Residual Structures

To enhance the model's nonlinear mapping capabilities and mitigate performance degradation with increasing model depth, each Transformer block incorporates a position-wise feed-forward network (FFN) followed by layer normalization.

$$\text{FFN}(x) = \max(0, x\mathbf{W}_1 + b_1)\mathbf{W}_2 + b_2 \quad (21)$$

In the meantime, residual connections are used to ensure reliable gradient back-propagation to the earlier GRU layers. This mechanism addresses the vanishing gradient problem and guarantees end-to-end differentiability throughout the hybrid GRU-Transformer architecture during training.

3.4. GRU-Transformer Framework

The analytical ECM derivations reveal that topology switching can introduce abrupt voltage transients, current redistribution, and nonlinear dynamic disturbances, which are difficult to model accurately using conventional model-based estimation approaches.

Motivated by these observations, the proposed GRU-Transformer architecture combines temporal continuity learning and topology-sensitive feature extraction to improve SOC estimation robustness under dynamically reconfigurable operating conditions. In this integrated framework, GRU layers are initially used to capture the inherent physical continuity and temporal dynamics of charge variation. Subsequently, Transformer layers leverage the global self-attention mechanism to identify and model logical correlations and abrupt dependencies triggered by topological mode transitions. This synergistic combination enables robust, high-precision SOC estimation across diverse operating conditions and reconfiguration scenarios, as illustrated in Figure 9. From an electrochemical perspective, topology switching changes the equivalent current distribution paths and transient voltage response characteristics of the battery module, resulting in dynamically varying nonlinear state-transition behaviors. The proposed GRU-Transformer framework is therefore designed to jointly capture electrochemical temporal continuity and topology-induced transient perturbations.

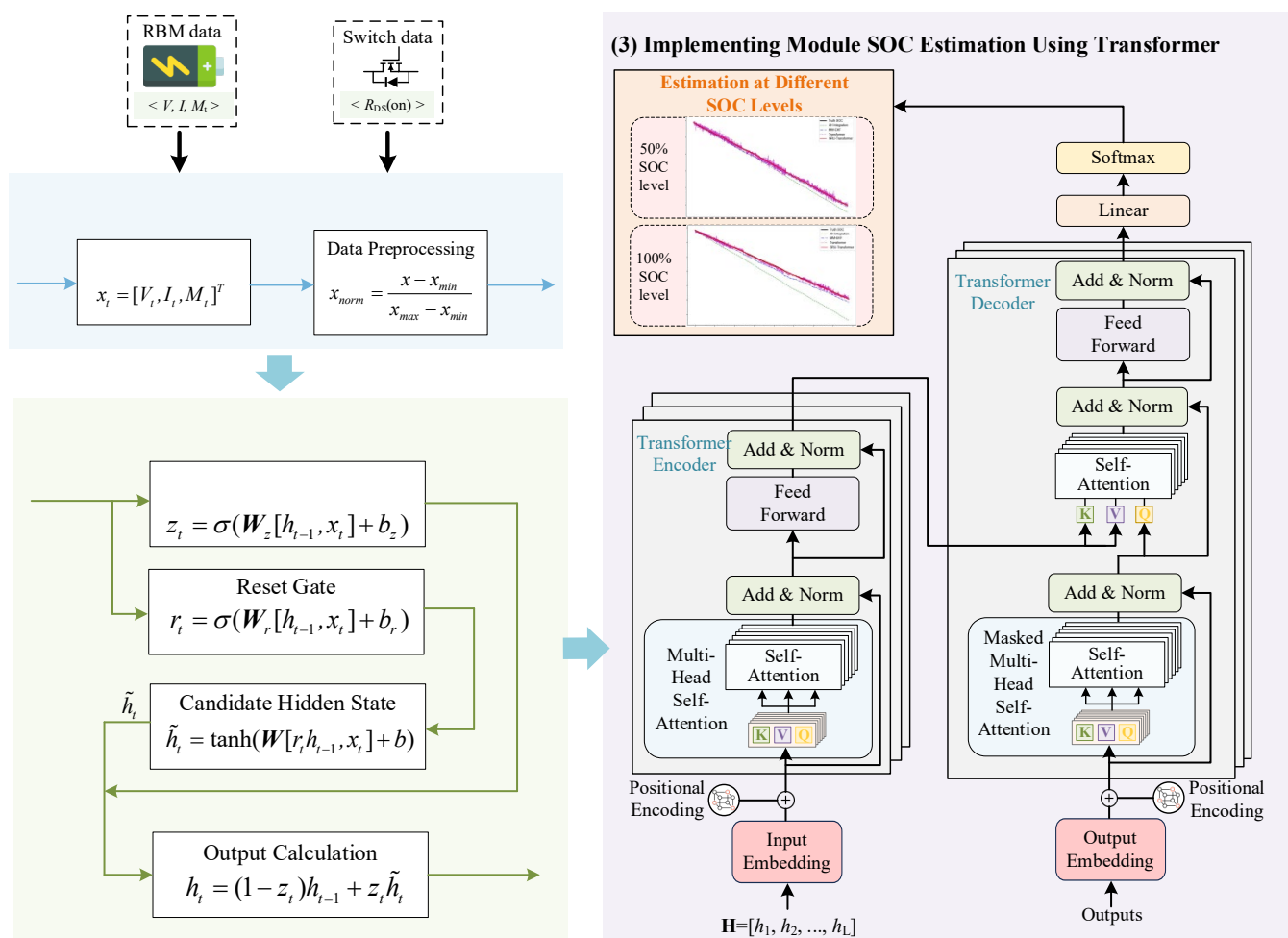


Figure 9. GRU-Transformer model architecture diagram.

It should be noted that the proposed framework adopts a relatively lightweight hybrid architecture by constraining the sequence length and controlling the network scale to reduce overall computational complexity. In particular, the GRU component is utilized to efficiently extract temporal continuity features and alleviate the long-sequence dependency modeling burden on the Transformer module, thereby improving computational efficiency and reducing unnecessary attention computation overhead. Compared with large-scale Transformer models commonly used in natural language processing tasks,

the Transformer encoder adopted in this work operates on low-dimensional battery state sequences, resulting in significantly reduced memory footprint and computational burden.

At the same time, the proposed architecture resolves signal discontinuities through a specialized division of labor: while the GRU preserves the inherent physical continuity of charge variation to prevent implausible SOC jumps, the Transformer leverages global self-attention to correlate topology flags with voltage transients. This synergy enables the model to distinguish between electrochemical dynamics and reconfiguration events, ensuring robust estimation during mode transitions.

In the proposed architecture, the GRU layer serves as the initial processing stage, directly receiving the multi-dimensional temporal input vector $\mathbf{x}_t$, which includes module terminal voltage, total current, and a discrete topology mode flag. The GRU's internal hidden state h_t acts as a dynamic integrator, continuously accumulating the effects of current over time through its gating mechanisms. This allows the model to maintain a physically consistent representation of the SOC evolution. During mode transitions in RBMs, which often cause abrupt discontinuities in observed terminal voltage, the GRU preserves continuity in the SOC trajectory by utilizing the previous hidden state h_{t-1} . As a result, the model effectively prevents physically implausible step changes or discontinuities in estimated SOC values.

The GRU output sequence is then fed into the Transformer encoder, where the self-attention mechanism performs a global analysis over the sliding window, paying special attention to shifts in the topology mode flag M_t . By capturing cross-correlations between module voltage V_t and topology mode flag M_t , specifically the topology-specific voltage variation signatures, the Transformer successfully distinguishes voltage changes caused by reconfiguration events from those induced by other dynamic operating conditions. It should be noted that the topology labels are only used to describe the external electrical configuration states of the RBM and do not contain direct SOC information or future-state reference information. Although the topology-switching logic follows predefined control rules, the corresponding transient electrical responses still exhibit strong nonlinearity and operating-condition dependency. Therefore, the model still needs to learn topology-dependent transient dynamic relationships from measured electrical signals rather than directly inferring SOC values from topology labels.

Overall, the proposed GRU-Transformer hybrid architecture achieves adaptive and high-precision SOC estimation across series, parallel, and hybrid series-parallel modes of the RBM through an end-to-end mapping mechanism. The GRU component primarily ensures steady-state estimation accuracy under all operating conditions by effectively capturing the continuous temporal dynamics of charge accumulation. Conversely, the Transformer component provides dynamic compensation during mode transitions, enabling the model to rapidly adapt to abrupt changes in electrical characteristics induced by topology reconfiguration. Moreover, the incorporation of residual connections ensures numerical stability and strong robustness against disturbances across dynamically varying topologies, guaranteeing both real-time computational performance and consistent estimation accuracy regardless of the prevailing connection mode. The overall implementation procedure of the proposed GRU-Transformer SOC estimation framework can be summarized as follows:

Step 1: Acquire battery operating data, including terminal voltage, current, temperature, and topology mode information under dynamically reconfigurable operating conditions.

Step 2: Construct the input state space by combining electrical measurements with the topology mode indicator M_t .

Step 3: Normalize and segment the sequential data into time-series samples for network training.

Step 4: Extract temporal continuity features through the GRU layer to preserve electrochemical state evolution characteristics.

Step 5: Apply the Transformer encoder with the MHSA mechanism to capture topology-dependent nonlinear transient features and signal discontinuities.

Step 6: Fuse the learned temporal and topology-sensitive representations to estimate the battery SOC through the fully connected output layer.

Step 7: Evaluate the estimation performance using MAE, RMSE, and maximum error metrics under different operating conditions.

The detailed algorithmic structure and implementation procedure of the GRU-Transformer hybrid model are presented in Algorithm 1.

Algorithm 1. Procedural Description of GRU-Transformer for SOC Estimation

```

1   Input: Input sequence  $\mathbf{x}_t \in \mathbb{R}^{L \times 3}$ , window size  $L$ ;
2   Initialize: GRU weights  $\mathbf{W}_{gru}$ , Transformer heads  $h$ , Linear layers  $\mathbf{W}_{fc}$ ,  $d_{model}$ ;
3   for each time step  $t$  in sliding window do
4        $h_t = \text{GRU}(\mathbf{x}_t, h_{t-1})$ ;
5       Store hidden state  $h_t$  in sequence  $\mathbf{H} = [h_1, h_2, \dots, h_L]$ ;
6   end
7    $\mathbf{H}_{pos} = \mathbf{H} + \text{Positional\_Encoding}(L, d_{model})$ ;
8    $\mathbf{Q} = \mathbf{H}_{pos} \mathbf{W}^Q$ ,  $\mathbf{K} = \mathbf{H}_{pos} \mathbf{W}^K$ ,  $\mathbf{V} = \mathbf{H}_{pos} \mathbf{W}^V$ ;
9    $\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}})\mathbf{V}$ ;
10   $\mathbf{Z} = \text{LayerNorm}(\text{Attention} + \mathbf{H}_{pos})$ ;
11   $\mathbf{Z}_{out} = \text{LayerNorm}(\text{FFN}(\mathbf{Z}) + \mathbf{Z})$ ;
12   $\mathbf{z}_{final} = \mathbf{Z}_{out}[L]$ ;
13   $\text{SOC}_{pred} = \mathbf{z}_{final} \mathbf{W}_{fc} + \mathbf{b}_{fc}$ ;
14  Output: Estimated SOC value.

```

4. Results and Discussion

This section presents experimental validation and a detailed analysis of SOC estimation performance for a four-cell RBM. The main goals are to clarify the impact of switching devices and verify the effectiveness of the proposed GRU-Transformer architecture. To thoroughly evaluate the model's robustness and adaptability across different electrochemical states, experiments are conducted at two representative initial SOC levels: 100% (fully charged) and 50% (half charged). It should be noted that the four-cell RBM platform adopted in this work is intended as an initial controllable validation platform for investigating topology-switching-induced transient behaviors and signal discontinuities under dynamically reconfigurable operating conditions. Since the proposed framework is topology-aware and fundamentally data-driven, it can theoretically be extended to larger battery arrays by incorporating higher-dimensional topology-state representations and distributed battery feature information.

4.1. Experimental Setting

In this study, the RBM is constructed using Samsung ICR18650-22F lithium-ion cells (SAMSUNG, Seremban, Malaysia), whose key specifications are outlined in Table 1: a nominal capacity of 2200 mAh, a nominal voltage of 3.6 V, and charge/discharge cut-off voltages of 4.2 V and 2.8 V, respectively. To simulate realistic load fluctuations and operating condition variations during system operation, the DST profile depicted in Figure 10 is

employed. This profile is used to assess the charging and discharging behavior of the RBM and to validate the proposed SOC estimation performance at the module level. In the DST current profile, positive values indicate discharging, while negative values indicate charging.

Table 1. Key specifications of the cell.

Cell Type	Nominal Capacity (mAh)	Nominal Voltage (V)	Charging Cut-Off Voltage (V)	Discharging Cut-Off Voltage (V)
ICR18650-22F	2200	3.6	4.2	2.8

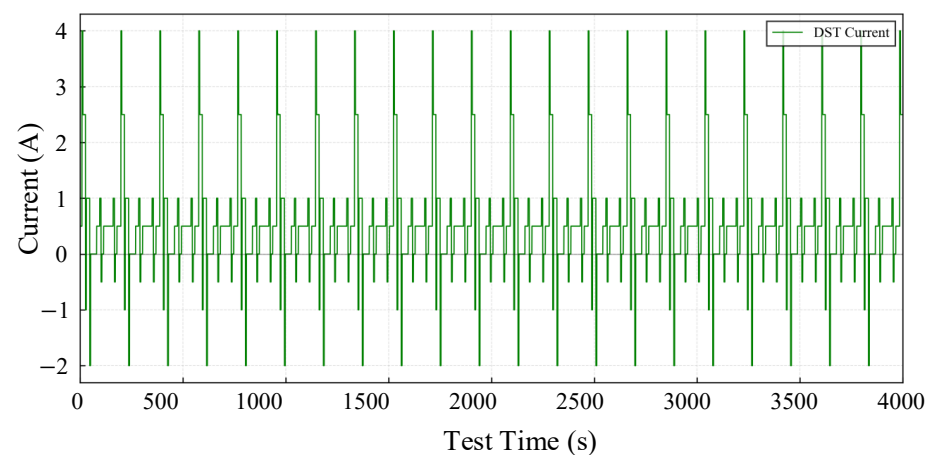


Figure 10. Dynamic test profiles of DST.

To quantitatively evaluate the SOC estimation performance of the proposed GRU-Transformer hybrid architecture for RBMs, this study employs two widely recognized error metrics, as defined in Equation (22): MAE and the Root Mean Square Error (RMSE).

$$\begin{cases} \text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \\ \text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \end{cases} \quad (22)$$

where $\hat{y}$ and y denote the model-predicted values and the reference values of SOC for the samples, respectively; n represents the total number of test samples.

All simulations and model implementations described in this study were developed and executed using PyTorch v2.3.1 under Python v3.12 on a Windows 11 operating system. Computations were performed on a personal computer equipped with an Intel Core i7-14650HX processor (Intel Corporation, Santa Clara, CA, USA) and an NVIDIA GeForce RTX 4060 GPU (NVIDIA, Santa Clara, CA, USA). Real battery experimental data were used for both training and evaluating the proposed GRU-Transformer model. The data acquisition process covered multiple charge/discharge cycles and different operating stages to capture transient electrical behaviors and topology-induced signal discontinuities. A total of 159,600 samples were employed, with the sampling frequency set at 1 Hz. Subsequently, the dataset was divided into training, validation, and testing subsets with ratios of 70%, 15%, and 15%, respectively. To improve the reliability of the generalization evaluation, the testing dataset was partitioned using operating segments mutually independent from the training process, thereby minimizing potential data leakage. In addition, all comparative methods adopted identical dataset partitioning conditions to ensure fairness and consistency in

the experimental comparisons. Table 2 summarizes the main configuration details of the experimental dataset.

Table 2. Experimental dataset configuration.

Parameter	Value
Number of samples	159,600
Sampling frequency	1 HZ
Training dataset ratio	70%
Validation dataset ratio	15%
Testing dataset ratio	15%

For the RBM, the initial topology was set to a parallel configuration. During the DST cycle, topology reconstruction was deliberately triggered at fixed intervals, forcing the system to switch among parallel, series, and hybrid series–parallel configurations. This experimental setup enables direct verification that the proposed method can effectively and robustly handle SOC estimation of the RBM under dynamic topology changes.

4.2. RBM State Estimation at 50% SOC Level

To meet the initial experimental conditions, the RBM, consisting of four lithium-ion cells, was first adjusted to a static equilibrium state at a 50% SOC level. At this SOC level, the OCV of lithium-ion batteries shows minimal variation with changes in SOC, resulting in a terminal voltage that is largely insensitive to SOC fluctuations. Under these conditions, the proposed estimation framework is required to reliably distinguish between transient voltage drops caused by topological reconfiguration and genuine electrochemical polarization effects, despite minimal background voltage fluctuations and strong coupling among input features. This setup provides a rigorous test of the model’s sensitivity and discrimination capability under challenging operating conditions.

For a rigorous comparative analysis of estimation performance under DST conditions, this study benchmarks the proposed GRU-Transformer hybrid method against three established approaches: the traditional AH integration method, the Interacting Multiple Model Extended Kalman Filter (IMM-EKF), and a pure Transformer-based model. All four methods are applied to estimate the SOC of the RBM. The selected benchmark methods are intended to represent conventional engineering estimation approaches (ampere-hour integration), advanced nonlinear filtering methods (IMM-EKF), and modern deep learning sequence modeling frameworks (Transformer), respectively. Therefore, the comparative analysis is designed to evaluate the proposed framework from both conventional engineering and modern data-driven perspectives. For fairness consideration, all comparative methods were evaluated under identical datasets, topology-switching conditions, sampling intervals, and operating profiles. The IMM-EKF parameters, including process-noise covariance and measurement-noise covariance matrices, were iteratively adjusted to improve adaptability under topology-switching-induced voltage transients. The Transformer baseline adopted a network scale and sequence configuration comparable to the proposed framework to ensure a relatively fair deep learning comparison. To facilitate an intuitive and direct comparison of their performance, the SOC trajectories obtained from the AH integration method, IMM-EKF, pure Transformer model, and the proposed GRU-Transformer architecture are plotted alongside the true SOC curve, as presented in Figure 11.

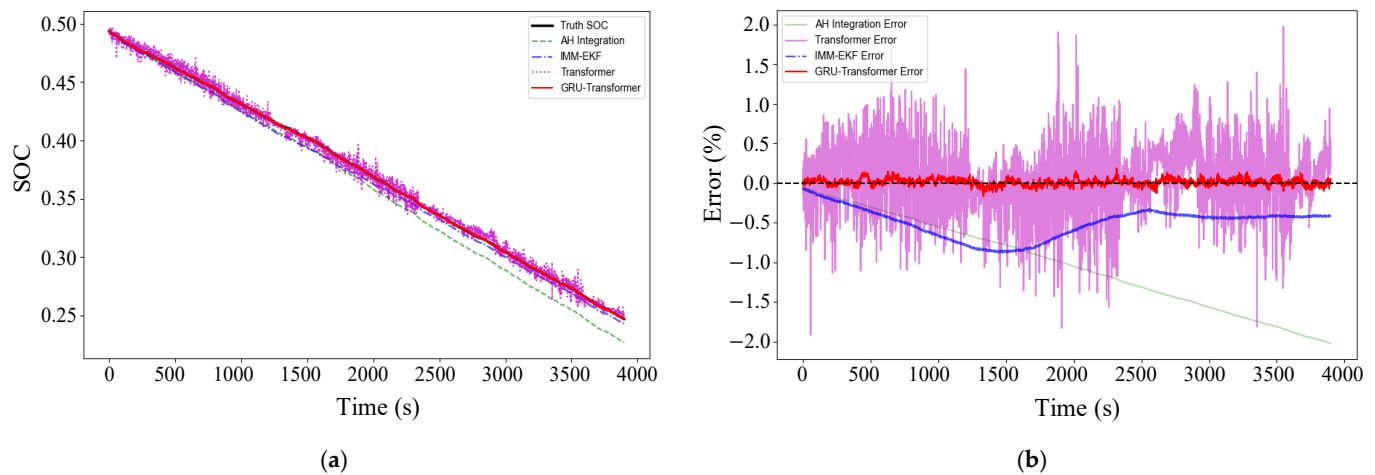


Figure 11. Estimated effect at 50% SOC level. (a) SOC estimation results. (b) SOC estimation error.

Figure 11a presents a comparison of the SOC estimation trajectories obtained from the AH integration method, IMM-EKF algorithm, standalone Transformer model, proposed GRU-Transformer hybrid architecture, and the true SOC. The AH integration method exhibits significant cumulative drift over time, primarily due to the lack of closed-loop feedback. The IMM-EKF algorithm outperforms the AH method but remains somewhat sluggish near dynamic switching points. Although the standalone Transformer is capable of capturing nonlinear dependencies, it shows noticeable numerical oscillations in the plateau region. In contrast, the proposed GRU-Transformer hybrid consistently achieves the highest agreement with the reference SOC trajectory. These results demonstrate that the synergistic combination of GRU's temporal memory capability and the Transformer's global self-attention mechanism effectively overcomes the estimation challenges posed by the small voltage gradients characteristic of the plateau region. This advantage is further evidenced in Figure 11b.

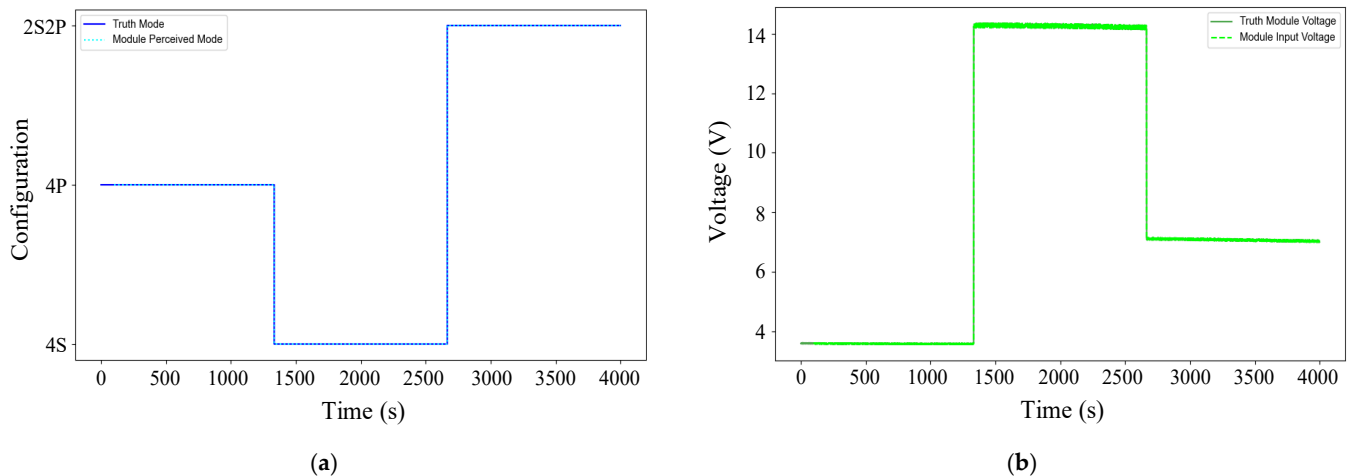
As illustrated in Figure 11b, the estimation error of the proposed GRU-Transformer remains consistently below 0.2% throughout the test, indicating excellent accuracy. In comparison, the standalone Transformer exhibits substantially larger error fluctuations, with peak deviations approaching 2%, highlighting its greater sensitivity to sensor noise and dynamic operating conditions. The markedly superior performance of the hybrid model also confirms its enhanced robustness in mitigating transient disturbances induced by RBM topology switching.

A quantitative comparison of SOC estimation performance across different algorithms is presented in Table 3. The proposed GRU-Transformer hybrid method exhibits clear superiority in both RMSE and MAE, achieving values of 0.044% and 0.035%, respectively. Compared to the traditional AH integration method and the IMM-EKF algorithm, the proposed approach reduces estimation errors by approximately an order of magnitude. It also offers nearly a tenfold improvement in accuracy over the standalone Transformer-based model. These significant precision gains strongly demonstrate the GRU-Transformer architecture's superior ability to model the complex nonlinear and topology-dependent dynamics inherent in RBMs.

To verify the accuracy and adaptability of the proposed method during dynamic topology reconfiguration in RBM operation, the internal connection topology of the RBM was periodically reconfigured under DST conditions. The mode identification and corresponding module voltage matching throughout the process were then evaluated, with results shown in Figure 12.

Table 3. SOC estimation error analysis of different algorithms at 50% SOC level.

Method	RMSE	MAE
GRU-Transformer	0.044%	0.035%
Transformer	0.419%	0.324%
IMM-EKF	0.536%	0.501%
AH Integration	1.179%	1.032%

**Figure 12.** Matching performance of GRU-Transformer at 50% SOC level. (a) Module voltage matching status. (b) Mode matching status.

As shown in Figure 12a, during the DST cycle, the RBM underwent three distinct topology transitions: from parallel to series and then to a hybrid series–parallel configuration. The proposed GRU-Transformer architecture successfully detected and accurately matched each mode change in real time. Figure 12b provides further confirmation of this capability. At around 1300 s, when the RBM switched from parallel to series configuration, the voltage features extracted and processed by the model aligned closely with the measured terminal voltage. Moreover, the observed voltage variations clearly reflected the underlying topology changes.

These results indicate that, through its MHSA mechanism, the GRU-Transformer effectively decouples voltage transients caused by topology switching from those arising from electrochemical dynamics. This decoupling significantly enhances the robustness of SOC estimation during reconfiguration transients.

4.3. RBM State Estimation at 100% SOC Level

The RBM was first adjusted to a 100% SOC level, corresponding to a fully charged condition, and the DST experiment was repeated to further evaluate the effectiveness of the proposed method. The performance of the GRU-Transformer hybrid architecture was compared against benchmark approaches under identical conditions. To meet the experimental initial conditions, the RBM was charged until it reached the full-charge cut-off state. At 100% SOC level, the open-circuit voltage exhibits a much higher $dV/dSOC$ slope compared to the plateau region near 50% SOC, resulting in a substantially improved signal-to-noise ratio. However, the pronounced polarization effects and complex electrochemical reactions characteristic of high-SOC regimes often cause large oscillations and estimation instability when conventional linear or model-based methods are used. The proposed GRU-Transformer architecture is specifically designed to address these challenges, enabling

stable, high-accuracy SOC estimation for RBMs even under such demanding operating conditions.

Under DST conditions, the SOC estimation results for the RBM derived from the evaluated algorithms are depicted in Figure 13. For a straightforward and intuitive performance comparison, the estimated SOC trajectories from all algorithms are plotted together in the same figure. Figure 13a illustrates the SOC estimation performance of the different methods, starting from 100% SOC. As the SOC declines from full charge to around 75%, the proposed GRU-Transformer hybrid architecture consistently exhibits the highest agreement with the true SOC trajectory. This superiority is further substantiated by the error comparison in Figure 13b. Unlike the significant and progressively growing deviations seen with the AH integration method and the IMM-EKF algorithm, the GRU-Transformer maintains its estimation error stably within $\pm 0.2\%$ throughout the test. In contrast, the standalone Transformer model displays marked high-frequency oscillations, with peak error values nearing 2%. These findings suggest that relying solely on the self-attention mechanism makes the model highly susceptible to voltage noise in high-SOC regions. At the same time, they underscore the excellent noise filtering capability and robustness of the GRU-Transformer hybrid against sensor noise and dynamic current fluctuations.

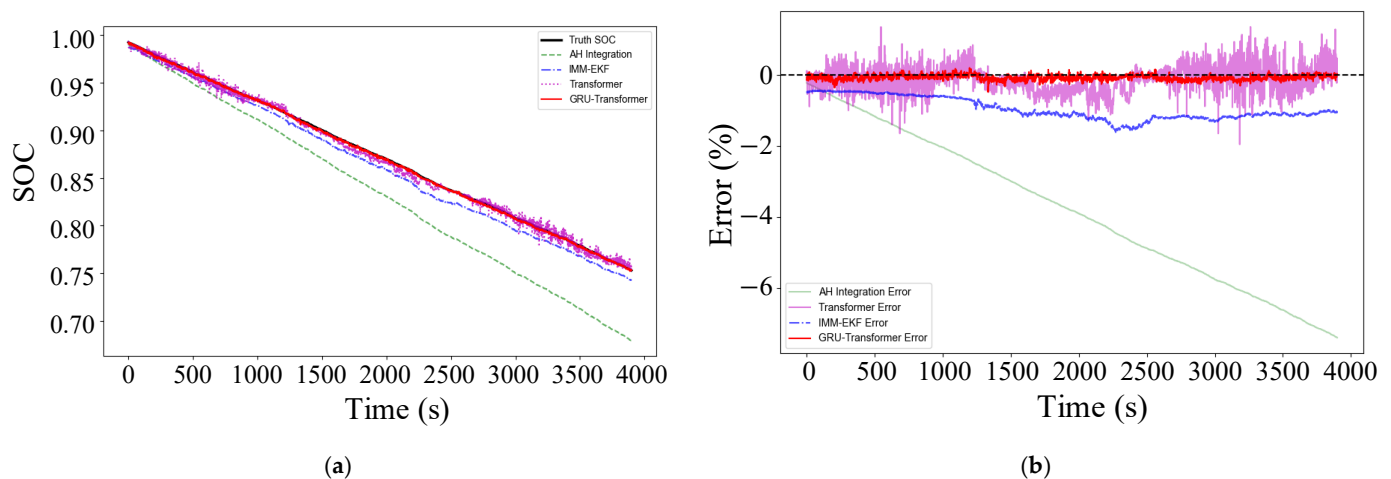


Figure 13. Estimated effect at 100% SOC level. (a) SOC estimation results. (b) SOC estimation error.

According to the results of the quantitative analysis, as listed in Table 4, the proposed GRU-Transformer method continues to show marked advantages in both RMSE and MAE metrics, achieving values of 0.103% and 0.085%, respectively. Although the errors are slightly higher than those at the 50% SOC level due to strong nonlinear polarization effects, this method still enhances accuracy by approximately 3.5 times compared to the Transformer model. Furthermore, compared to the IMM-EKF method, it achieves nearly an 11-fold improvement in accuracy. This demonstrates that the GRU-Transformer maintains strong robustness and generalization capability under conditions of maximum system stress and strongest nonlinearity, thereby meeting the safety estimation requirements for RBMs at full charge. The improved estimation accuracy is mainly attributed to the cooperative mechanism between the GRU temporal memory structure and the Transformer-based topology-sensitive feature extraction capability, which enables effective suppression of topology-switching-induced signal discontinuities.

As illustrated in Figure 14, the proposed GRU-Transformer model accurately identifies topological patterns and maintains a smooth error curve, even during abrupt terminal voltage jumps caused by mode switching. This result strongly confirms the effectiveness of the MHSA mechanism in precisely capturing topology mode indicators. It successfully

decouples the logic of system reconfiguration from the electrochemical state information of the battery cells. Furthermore, these findings demonstrate that the proposed method consistently delivers high-precision SOC estimation under complex and dynamically varying topologies.

Table 4. SOC estimation error analysis of different algorithms at 100% SOC level.

Method	RMSE	MAE
GRU-Transformer	0.103%	0.085%
Transformer	0.383%	0.304%
IMM-EKF	1.008%	0.958%
AH Integration	4.343%	3.820%

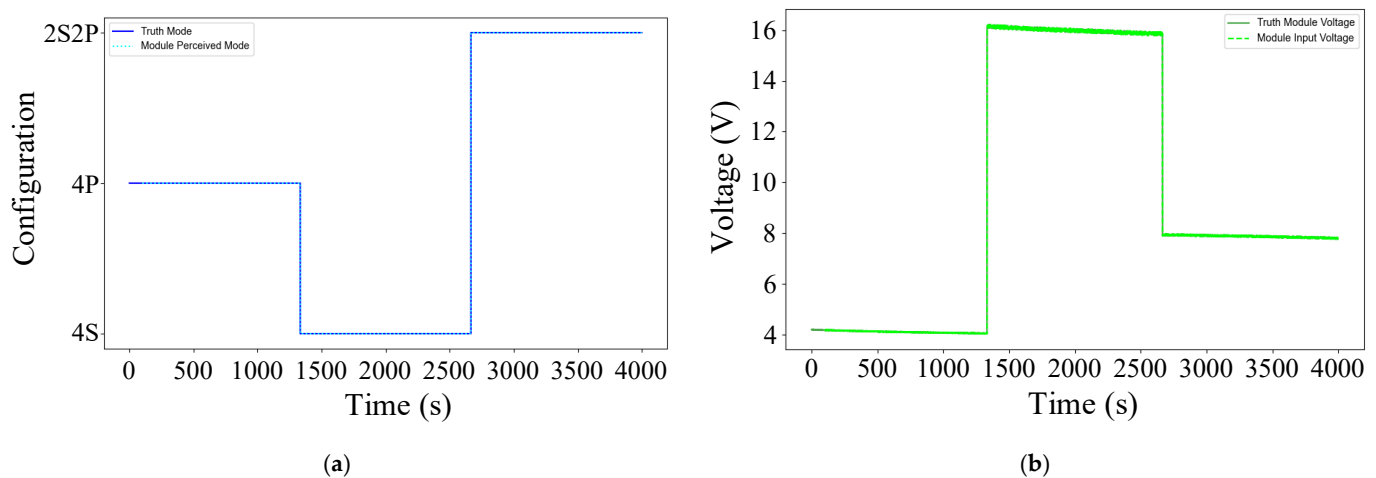


Figure 14. Matching performance of GRU-Transformer at 100% SOC level. (a) Module voltage matching status. (b) Mode matching status.

The experimental results indicate that topology-aware deep learning architectures can effectively improve SOC estimation robustness under dynamically reconfigurable operating conditions. This suggests that combining temporal continuity learning with topology-sensitive feature extraction provides a promising solution for handling strong non-stationary behaviors in next-generation intelligent battery management systems. Furthermore, the proposed framework demonstrates potential applicability in adaptive energy storage systems requiring real-time topology reconfiguration and fault-tolerant operation.

4.4. Cross-Cycle Generalization Validation

To further evaluate the generalization capability of the proposed topology-aware SOC estimation framework, additional validation experiments were conducted on the RBM under the Federal Urban Driving Schedule (FUDS) operating profile at the 100% SOC level, as illustrated in Figure 15. Although the current study primarily adopts the DST profile to investigate topology-switching-induced transient behaviors, further validation under standard driving cycles such as FUDS can improve the comprehensiveness of the generalization evaluation. This experiment was mainly designed to assess the adaptability of the proposed framework under previously unseen operating conditions. Compared with the DST profile, the FUDS cycle exhibits different load fluctuation characteristics and transient current distributions, thereby providing a more challenging validation scenario for topology-sensitive SOC estimation.

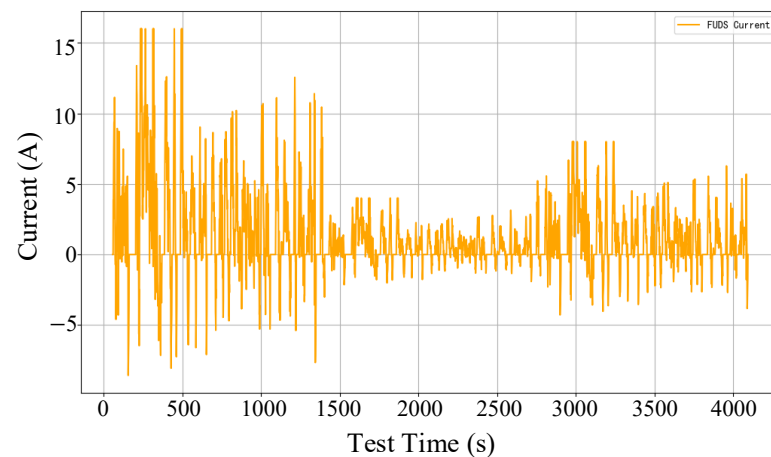


Figure 15. Dynamic test profiles of FUDS.

The experimental results demonstrate that, as shown in Figure 16 and Table 5, the proposed framework is still capable of maintaining stable estimation accuracy and satisfactory dynamic tracking capability under the FUDS operating condition, indicating good generalization performance across different driving-cycle profiles.

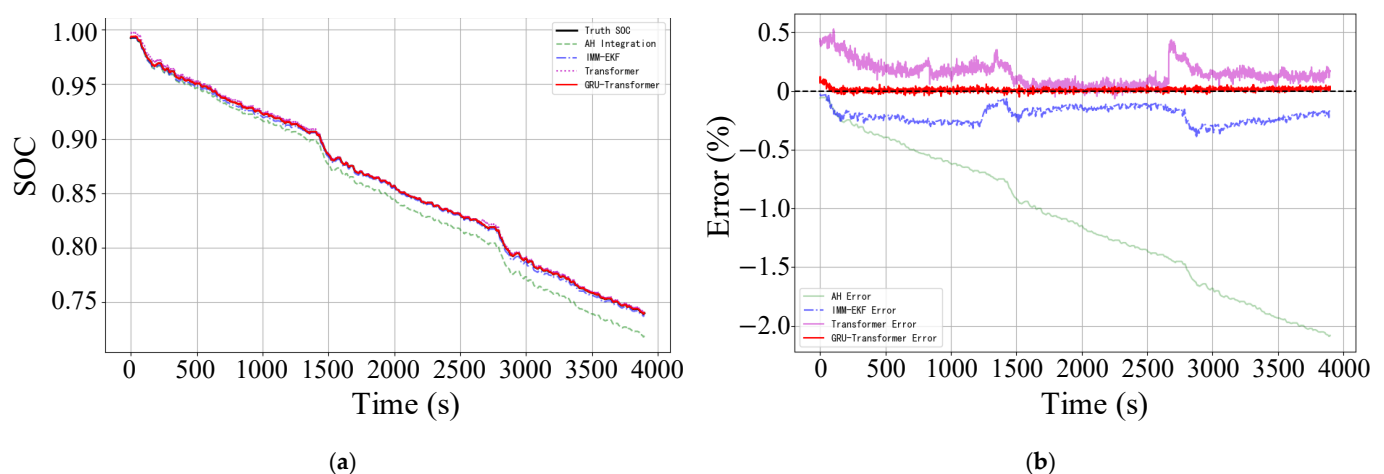


Figure 16. SOC estimation under the FUDS operating profile at the 100% initial SOC level. (a) SOC estimation results. (b) SOC estimation error.

Table 5. SOC estimation error analysis of different algorithms under the FUDS profile.

Method	RMSE	MAE
GRU-Transformer	0.018%	0.014%
Transformer	0.181%	0.152%
IMM-EKF	0.215%	0.205%
AH Integration	1.257%	1.117%

4.5. Discussion

The high-frequency estimation fluctuations observed in the experimental results are mainly associated with topology-switching-induced voltage discontinuities, transient current redistribution, and nonlinear coupling dynamics during topology-transition intervals. Due to the strong temporal sensitivity of sequence learning models to dynamic temporal features, abrupt topology-transition events may introduce localized transient estimation oscillations under dynamically reconfigurable operating conditions. Meanwhile, excessive high-frequency estimation oscillations may potentially affect downstream control modules

in closed-loop battery management systems, including balancing control, power allocation strategies, fault diagnosis, and topology management decisions. Nevertheless, the proposed GRU-Transformer framework still maintains stable overall estimation accuracy and effectively suppresses long-term estimation drift under dynamically reconfigurable operating conditions.

It should be noted that, in practical RBM applications, topology reconstruction may occur at relatively short time intervals due to dynamic balancing or fault-management requirements. Under such high-frequency switching conditions, rapid voltage transients and signal discontinuities can significantly increase estimation difficulty. Nevertheless, the proposed GRU-Transformer architecture maintains stable SOC estimation performance throughout repeated topology transitions during the DST cycle. The GRU layer preserves temporal continuity through hidden-state memory propagation, while the Transformer layer dynamically captures topology-dependent abrupt variations via the MHSA mechanism. Consequently, no noticeable estimation divergence or oscillatory instability is observed during successive switching events, demonstrating strong robustness and dynamic adaptability under highly non-stationary operating conditions.

The current experimental validation mainly focuses on topology-switching-induced dynamic behavior under simplified cell-consistency conditions. In practical RBM applications, cell-to-cell inconsistencies caused by aging, capacity degradation, thermal gradients, and parameter mismatch may further increase system nonlinearity and estimation difficulty. Since the proposed framework is fundamentally data-driven, it can theoretically be extended to such complex operating scenarios provided that representative training datasets are available. Experimental validation under realistic inconsistency conditions will be investigated in future work. Meanwhile, since the proposed GRU-Transformer framework is fundamentally data-driven, temperature measurements and thermal-state variables can be naturally incorporated into the input state space to enable electro-thermal coupled feature learning. This provides a potential pathway for extending the proposed framework toward temperature-aware SOC estimation under realistic operating conditions. Furthermore, large-scale industrial battery systems typically involve more complex electrical coupling behaviors, thermal interaction processes, and topology management challenges. Therefore, additional large-scale experimental validations are still required to further evaluate the scalability and engineering applicability of the proposed framework.

The current validation mainly focuses on representative medium- and high-SOC operating conditions. Although the proposed GRU-Transformer framework demonstrates stable estimation performance under these scenarios, additional validations under low, intermediate, and randomly initialized SOC conditions would further improve the comprehensiveness of the sensitivity analysis. Since the proposed method is fundamentally data-driven, its adaptability under different SOC intervals depends on the diversity and representativeness of the training dataset. Future work will therefore further investigate the robustness and generalization capability of the proposed framework under broader SOC operating ranges, including 25%, 75%, and random initial SOC conditions.

5. Conclusions

To address the limitations of existing state estimation methods, which primarily focus on individual battery cells and inadequately account for switching effects and dynamic topological variations, this paper proposes a GRU-Transformer-based SOC estimation algorithm tailored for reconfigurable battery modules. This approach enables high-precision SOC estimation during dynamic reconfiguration of modular battery systems. Experimental results show that the proposed GRU-Transformer method delivers outstanding performance across different SOC levels. It exhibits excellent robustness at both 50% SOC and

under high-voltage polarization at 100% SOC, with MAE values as low as 0.035% and 0.085%. This effectively overcomes the cumulative drift of conventional AH integration methods and the numerical oscillations of single deep learning models. More importantly, the GRU-Transformer can accurately recognize module topology patterns and maintain smooth estimation curves even during abrupt terminal voltage jumps due to reconfiguration, fully validating its ability to decouple physical logic from electrochemical states. The accurate and robust state estimation achieved by the proposed method effectively mitigates overcharge and overdischarge risks, thereby improving the safety and resilience of battery energy storage systems. Moreover, it can effectively cope with topological uncertainty and dynamic load changes and has high engineering application value.

It should be noted that the performance of the proposed GRU-Transformer framework is influenced by several key hyperparameters, including the GRU hidden-state dimension, input sequence length, and Transformer attention configuration. Among them, the sequence length and hidden-state dimension exhibit relatively high sensitivity, since they directly affect the extraction capability of temporal continuity features. The Transformer attention configuration mainly influences the model's ability to capture topology-dependent transient characteristics and signal discontinuities. Although larger network dimensions may enhance nonlinear representation capability, they can also increase computational burden and the risk of overfitting. Therefore, appropriate parameter selection is essential for achieving a balance among estimation accuracy, robustness, and computational efficiency.

Despite the promising SOC estimation performance demonstrated in this work, several limitations still exist. First, the current experimental validation is mainly conducted under relatively controlled operating conditions, and more comprehensive validations under realistic battery inconsistency scenarios are still required. Second, the equivalent circuit derivations adopt simplified assumptions of cell consistency to facilitate topology-dependent electrical analysis. In practical applications, battery aging, parameter mismatch, thermal imbalance, and asynchronous SOC evolution may introduce additional nonlinear uncertainties.

In future research, the robustness of the proposed framework will be further validated under real-world cell inconsistency conditions, including complex scenarios such as battery aging, parameter mismatch, thermal gradients, and asynchronous SOC evolution. In addition, as a data-driven deep learning framework, the proposed GRU-Transformer architecture still relies on the diversity and representativeness of the training dataset to achieve high estimation accuracy under dynamically reconfigurable operating conditions. Future work will therefore investigate transfer learning strategies to improve cross-platform adaptability and reduce the dependence on large-scale labeled training data. This is expected to enhance the applicability of the proposed method to different battery chemistries, module configurations, and operating environments while lowering retraining costs for practical deployment. Meanwhile, future work will further investigate electro-thermal coupled modeling and temperature-aware topology-sensitive SOC estimation methods under dynamically reconfigurable operating conditions, as well as embedded deployment optimization strategies for practical BMS hardware platforms. In addition, systematic quantitative ablation studies will be conducted to analyze the independent contributions of individual architectural components.

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